

[Malek Ahmad](#) | majobs112002@gmail.com | +972549498961 | [Linkedin](#) | [Github](#) | [Portfolio](#)

Python Developer experienced in **building scalable backend systems** using **Flask and RESTful APIs**, with solid knowledge of **databases and backend architecture**. Strong foundation in **machine learning and AI**, combining analytical thinking with practical system design. A **fast and self-driven learner** with **strong problem-solving skills**, **motivated team player**, and able to **work effectively under pressure**.

Skills

Programming languages: C, C++, Java, Python, JavaScript.

Operating Systems: Linux, Windows.

Backend: Flask, REST APIs, Authentication, Scalable Architecture, Node.js.

Database: SQL (MySQL), Firestore, MongoDB.

IDEs: CLion, PyCharm, VS Code, Cursor.

Education

2021 –2025: Azrieli College of engineering | Software engineering | GPA 90.83

Courses: Data Structures and Algorithms | Operating Systems | Database | Computer Networks | Distributed Systems | Introduction to machine learning | Reinforcement learning.

Projects

- **2025: Retrieval-Augmented Generation (RAG) System for Multilingual Document Search**, [Code link](#)

Built a scalable **RAG web application** for internal use in academic institutions, enabling accurate Q&A over large, interconnected document collections in **Hebrew, Arabic, and English**.

Developed the **backend using Python (Flask), LangChain, and FAISS**. Designed a **React-based frontend** with a chat interface and file upload system.

- **2025: Reinforcement Learning AI for SOS Game**, [Code link](#)

Developed a self-learning AI agent for the SOS game **using Monte Carlo Tree Search (MCTS), Puct and deep learning**. Built **using Python, TensorFlow/Keras**.

- **2025: Parallel Image Matching System**, [Code link](#)

Implemented a hybrid **parallel image-matching system** in **C** using **MPI** for distributed multi-process execution and **OpenMP** for intra-process thread parallelism, with **dynamic workload** distribution across cluster nodes.

- **2024: Unsupervised Learning Pipeline – K-Means & Gaussian Mixture Models**, [Code link](#)

Built an **unsupervised learning pipeline** using **K-Means and GMM**, applying **PCA** for efficient clustering and analysis. Compared models using **ARI/accuracy**, focusing on **convergence, stability, and performance trade-offs**.

Certificates

- **2025: Place-IL Developers Internship Program**

- Successfully passed a **competitive** selection process (**250+ candidates**), including **coding challenges** in **OOP, data structures, and databases**.
- Successfully completed **HR, technical, and group** interview stages.

- **2024: CS50: Introduction to Artificial Intelligence with Python – Harvard University (edX)** [Projects and Certificate](#).

- Developed practical projects such as **image classification, sentiment analysis, Sudoku solver, and reinforcement learning agents**.
- Gained hands-on experience in **AI concepts**, including search algorithms, knowledge representation, optimization, and machine learning.

Experience

2022 – 2025: Tutor | Azrieli College of Engineering | Jerusalem

Tutored **4-5 students per year** in **programming, algorithms, math** and **physics** courses, improving my responsibility, self-confidence, and socialization skills.

Languages

Arabic (Native), Hebrew (Fluent), English (High level)